ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** autotrophic	Plants make their own food.
2	**(c)** bile	Bile juice.
3	**(d)** four	Two atria and two ventricles.
4	**(a)** CO_2 and water	CO_2 + H_2O.
5	**(a)** liver	Liver.
6	**(b)** arteries	Arteries carry blood away from the heart.
7	**(d)** chlorophyll	Chlorophyll.
8	**(c)** small intestine	Intestinal juice.
9	**(c)** RBC	Red blood cells.
10	**(d)** oxygen	O_2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Process of obtaining and using food (1); Autotrophic and heterotrophic (1).	1+1
12	Rhythmic contraction of muscles of the alimentary canal (1); Pushes food forward (1).	1+1
13	Loss of water as vapour through stomata (1); From aerial parts of the plant (1).	1+1
14	Carbon dioxide + water (1); gives glucose + oxygen in presence of light and chlorophyll (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Carbon dioxide from air (1); Water from soil (1); Light from Sun, chlorophyll in chloroplast (1).	1+1+1
16	Absorbs water (1); Stores undigested food (1); Egestion (1).	1+1+1
17	Stored in vacuoles (1); Falling leaves (1); Gums, resins, latex (1).	1+1+1
	OR: Holozoic: humans (1); Saprophytic: fungi (1); Parasitic: Cuscuta (1).	1+1+1
18	Breathing: inhale and exhale air (1); Respiration: release of energy from food in cells (1); Breathing is a physical process (1).	1+1+1
	OR: Absorb water and minerals (1); Increase surface area (1); Present in roots (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Takes place in chloroplast (1); Chlorophyll absorbs light (1); Water splits (1); CO_2 is reduced to glucose (1); $6\,CO_2 + 12\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2 + 6\,H_2O$ (1).	1+1+1+1+1
	OR: Xylem: water; phloem: food (1); Xylem: dead cells; phloem living (1); Xylem upward; phloem both ways (1); Xylem passive; phloem active (1); Vessels/tracheids vs sieve tubes (1).	1+1+1+1+1
20	Ingestion (1); Digestion (1); Absorption (1); Assimilation (1); Egestion (1).	1+1+1+1+1
	OR: Aerobic: glucose to CO_2 + H_2O + energy (1); Yeast: glucose to ethanol + CO_2 (1); Muscle: glucose to lactic acid (1); Energy stored in ATP (1); Anaerobic yields less energy (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that plants respire only at night and photosynthesise only by day; plants respire all the time and photosynthesise in the presence of light.
- Mixing up xylem and phloem: xylem carries water and minerals (one-way, upwards); phloem carries food (both directions).
- Saying pulmonary artery carries oxygenated blood; the pulmonary **artery** carries deoxygenated blood to the lungs and the pulmonary **vein** carries oxygenated blood back.
- Confusing breathing with respiration; breathing is gas exchange, respiration is the breakdown of glucose in cells to release energy.
- Stating that pyruvate is formed in the mitochondria; glucose is broken down to pyruvate in the cytoplasm.
- Believing the kidney makes urine only by filtration; selective reabsorption and the amount of water excreted are equally important steps.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** villi	Villi in small intestine.
2	**(d)** platelets	Platelets.
3	**(a)** leaves	Leaves.
4	**(c)** large intestine	Large intestine.
5	**(d)** kidneys	Kidneys.
6	**(a)** pseudopodia	Finger-like extensions.
7	**(a)** gall bladder	Gall bladder.
8	**(c)** nephron	Nephron.
9	**(a)** bread mould	Saprophyte.
10	**(c)** stomach	Gastric glands.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Arteries (1); Veins and capillaries (1).	1+1
12	Xylem: transport water and minerals (1); Phloem: transport food (1).	1+1
13	Tiny pores on leaves (1); Gas exchange and transpiration (1).	1+1
14	Release of energy from food (1); Aerobic and anaerobic (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Salivary: amylase (1); Liver: bile (1); Pancreas: pancreatic juice (1).	1+1+1
16	Aerobic needs oxygen (1); Anaerobic does not (1); More energy in aerobic (1).	1+1+1
17	Blood passes through heart twice (1); Pulmonary and systemic (1); Present in humans (1).	1+1+1
	OR: Moistens food (1); Amylase digests starch (1); Helps in swallowing (1).	1+1+1
18	Arteries carry blood from heart (1); Veins have valves (1); Arteries have thicker walls (1).	1+1+1
	OR: Adenosine triphosphate (1); Energy currency of cells (1); Releases energy on breaking (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mouth (1); Oesophagus, stomach (1); Small intestine, large intestine (1); Liver, pancreas (1); Rectum, anus (1).	1+1+1+1+1
	OR: Amoeba: pseudopodia (1); Food vacuole (1); Paramecium: cilia sweep food to a specific spot (1); Digestion in vacuole (1); Waste expelled (1).	1+1+1+1+1
20	Nostrils, pharynx, larynx (1); Trachea and bronchi (1); Bronchioles and alveoli (1); Diaphragm and ribs (1); Gas exchange at alveoli (1).	1+1+1+1+1
	OR: Removal of waste from blood artificially (1); Blood passes through tubes in dialysis fluid (1); Urea and wastes diffuse out (1); Cleaned blood returns (1); Used for kidney failure (1).	1+1+1+1+1

Mistakes to avoid:

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- Mixing up xylem and phloem: xylem carries water and minerals (one-way, upwards); phloem carries food (both directions).
- Saying pulmonary artery carries oxygenated blood; the pulmonary **artery** carries deoxygenated blood to the lungs and the pulmonary **vein** carries oxygenated blood back.
- Confusing breathing with respiration; breathing is gas exchange, respiration is the breakdown of glucose in cells to release energy.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** xylem	Xylem.
2	**(a)** parasite	It is parasitic.
3	**(b)** peristalsis	Rhythmic contractions.
4	**(d)** phloem	Phloem.
5	**(d)** amylase	Salivary amylase.
6	**(c)** haemoglobin	In RBCs.
7	**(b)** transpiration	Transpiration.
8	**(b)** pepsin	Pepsin digests proteins.
9	**(d)** alveoli	Alveoli.
10	**(b)** red blood cells	RBC.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Glomerulus and Bowman's capsule (1); Tubule and collecting duct (1).	1+1
12	Breakdown of complex food into simple (1); Salivary amylase (1).	1+1
13	Removal of nitrogenous wastes (1); From the body of organisms (1).	1+1
14	Contain haemoglobin (1); Carry oxygen (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Churns food (1); HCl and pepsin act on proteins (1); Mucus protects the lining (1).	1+1+1
16	Transports oxygen (1); Transports food and waste (1); Fights infection and clots (1).	1+1+1
17	Finger-like projection (1); Used for movement (1); Used for engulfing food (1).	1+1+1
	OR: Complete digestion (1); Absorption by villi (1); Receives bile and pancreatic juice (1).	1+1+1
18	Kidneys: filter blood (1); Ureters: carry urine (1); Bladder stores; urethra releases (1).	1+1+1
	OR: Large surface area (1); Thin walls (1); Rich in capillaries (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Right atrium (1); Right ventricle to lungs (1); Left atrium (1); Left ventricle to body (1); Valves ensure one-way flow (1).	1+1+1+1+1
	OR: Xylem carries water; phloem food (1); Xylem upward; phloem both directions (1); Xylem: dead; phloem: living (1); Xylem: vessels; phloem: sieve tubes (1); Xylem uses transpiration pull; phloem uses ATP (1).	1+1+1+1+1
20	Blood enters glomerulus (1); Filtration (1); Selective reabsorption (1); Urine in collecting duct (1); Stored in bladder (1).	1+1+1+1+1
	OR: Bile: emulsifies fat (1); HCl: acid medium (1); Amylase: starch (1); Pepsin: proteins (1); Lipase: fats (1).	1+1+1+1+1

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